

METODE INTELIGENTE DE REZOLVARE A PROBLEMELOR REALE



Laura Dioşan
Image segmentation

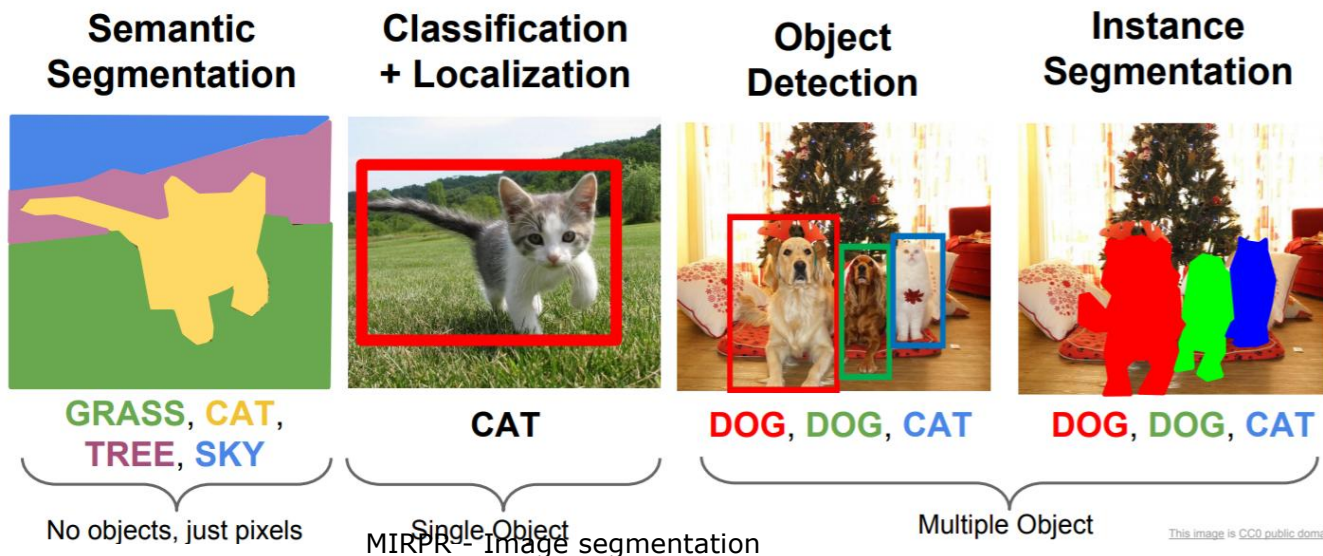
Automatic image processing

□ Image classification



CAT or DOG
or OTHER?

□ Other tasks



Automatic image processing

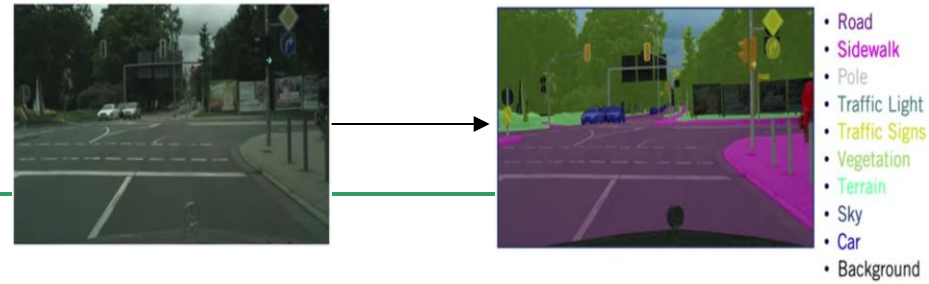
□ Image classification

- Does an image contain object X? [yes/no]

□ Image detection and segmentation

- Does an image contain object X? [yes/no]
- Where is the object X? → Location of the object
 - Pixel-based granularity → semantic/instance segmentation
 - Object-based granularity → object detection
- Which object does this image contain? [where?]
- Aprox. localisation (Bounding box)
- Accurate localisation (contour) → Segmentation

Image segmentation



□ Problem

■ Aim

- Classify each pixel

■ Tasks

- How many segments?
- How many objects in an image?

Image segmentation

□ Problem → Tasks

■ Semantic segmentation

- Labels for every pixel
- No differences across different instances of the same object

■ Instance segmentation

- Labels for every pixel
- unique label to every instance of a particular object in the image
- Special topic: Panoptic segmentation
 - Instance segmentation for background

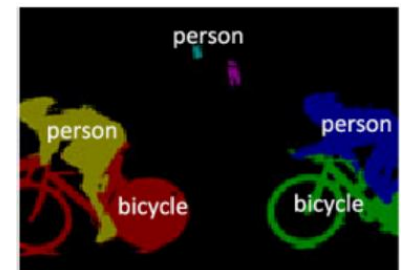
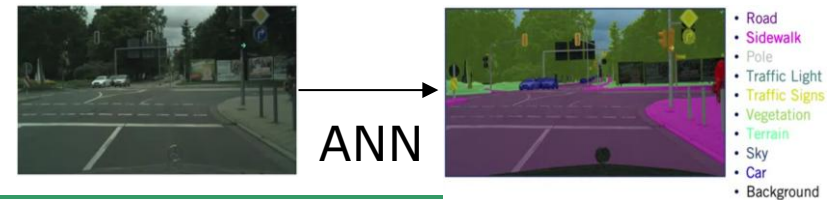


Image segmentation



□ Problem

■ Challenges

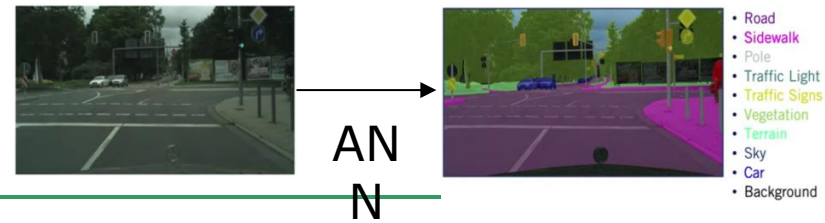
- Occlusion, Truncation, Scale, Illumination
- Smooth boundaries

■ Evaluation

- TP - #pixels correctly classified as belonging to class X
- FP - #pixels classified as belonging to class X, but they belong to other classes
- FN - #pixels that belong to class X, but are not classified as belonging to class X
- $IOU_{class} = TP / (TP + FP + FN)$ – over all images

Ground Truth	Prediction	Class: Road	Class: Sidewalk																		
<table><tr><td>R</td><td>R</td><td>R</td></tr><tr><td>R</td><td>R</td><td>S</td></tr><tr><td>S</td><td>S</td><td>S</td></tr></table>	R	R	R	R	R	S	S	S	S	<table><tr><td>S</td><td>R</td><td>S</td></tr><tr><td>R</td><td>R</td><td>S</td></tr><tr><td>S</td><td>S</td><td>S</td></tr></table>	S	R	S	R	R	S	S	S	S	$TP = 3$ $FP = 0$ $FN = 2$	$TP = 4$ $FP = 2$ $FN = 0$
R	R	R																			
R	R	S																			
S	S	S																			
S	R	S																			
R	R	S																			
S	S	S																			
		$IOU_{Road} = \frac{3}{3+0+2} = \frac{3}{5}$	$IOU_{Road} = \frac{4}{4+2+0} = \frac{4}{6}$																		

Image segmentation



□ Problem

■ Datasets

□ 2001 Berkeley

- <https://www2.eecs.berkeley.edu/Research/Projects/CS/vision/bsds/>

- Good for edge detection problem (also)

□ 2005 – Pascal VOC

- 20 classes

□ 2015 – COCO dataset (detection and segmentation)

- <https://cocodataset.org/#detection-2015>

- 91 classes

□ 2015 – CityScapes

- <https://www.cityscapes-dataset.com/>

- 30 classes grouped in 8 categories

□ CamVid

- <http://mi.eng.cam.ac.uk/research/projects/VideoRec/CamVid/>

Image segmentation

□ How?

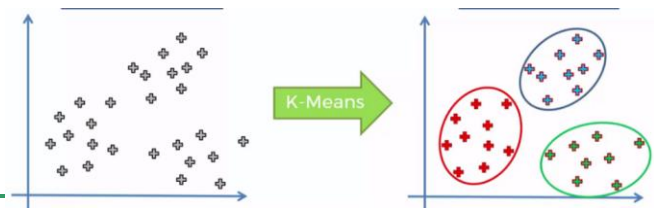
■ Before Computer Vision

- Gestalt: whole or group
 - Whole is greater than sum of its parts
 - Relationships among parts can yield new properties/features
- Psychologists identified series of factors that predispose set of elements to be grouped (by human visual system)
 - "I stand at the window and see a house, trees, sky. Theoretically I might say there were 327 brightnesses and nuances of colour. Do I have "327"? No. I have sky, house, and trees." Max Wertheimer (1880-1943)

■ Computer Vision's era

- Segmentation as clustering (K-means, GAMMs and EM, Mean Shift, ...)
- Segmentation as grouping by boundaries
- Graph-based segmentation
- Segmentation as energy minimization
- Region-based segmentation (->Thresholding, Region growing)
- Edge detection segmentation
- Deep learning algorithms

Image segmentation



□ How? -> Computer Vision's era

■ Segmentation as clustering

□ Main idea

- Group the "similar" pixels into clusters

□ Algorithms:

- K-means, GAMMs and EM, Mean Shift, ...

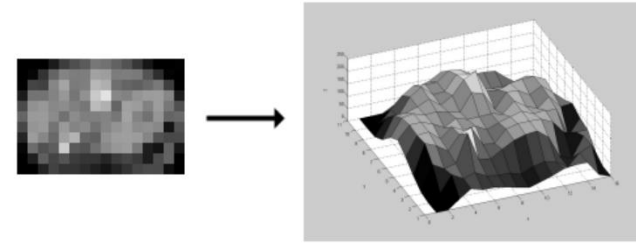
- <https://scikit-learn.org/stable/modules/clustering.html>

□ See

- Comaniciu, D., & Meer, P. (2002). Mean shift: A robust approach toward feature space analysis. *IEEE Transactions on pattern analysis and machine intelligence*, 24(5), 603-619. <https://courses.csail.mit.edu/6.869/handouts/PAMIMeanshift.pdf>
- <http://cs229.stanford.edu/notes2020spring/cs229-notes8.pdf>
- <http://cs229.stanford.edu/notes2020spring/cs229-notes7b.pdf>

- + works well on a small dataset with convex clusters
- - large computational time, shape of clusters

Image segmentation



□ How? -> Computer Vision's era

■ Segmentation as grouping by boundaries

□ Main idea

- Edge-based methods

□ Algorithms:

- Watershed – good for hierarchical segmentation

- the image is regarded as a topographic landscape with ridges and valleys

- Level-sets

□ See

- <https://members.accu.org/index.php/journals/1469>
- https://hub.gke2.mybinder.org/user/scikit-image-scikit-image-lpeqi3jb/notebooks/notebooks/auto_examples/segmentation/plot_watershed.ipynb

- + Fast (apply filters)

- - if there are too many edges or less contrast objects

Image segmentation

□ How? -> Computer Vision's era

■ Graph-based segmentation

□ Main idea

- Images as graphs (nodes – pixels, weights (affinity matrix) – location/intensity/color/textureFilters) and break graph in segments

□ Algorithms

- Graph-Cut – eigen values of affinity matrix
- Min-cut

□ See

- <http://cs.brown.edu/people/pfelzens/segment/>
- Shi, J., & Malik, J. (2000). Normalized cuts and image segmentation. *IEEE Transactions on pattern analysis and machine intelligence*, 22(8), 888-905.
https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=868688&casa_token=b23BGFY2CwAAAAA:wbUB6ZhAc3vHP11li6cl2NyjfpI0vAHGefdvKegPJLacEiB332Xn0EnIF94R1qKk4MUdXgcFALPA&tag=1

- + Flexible to choice of affinity matrix
- + Generally works better than other methods
- - Can be expensive, especially with many cuts.
- - Bias toward balanced partitions
- - Constrained by affinity matrix model

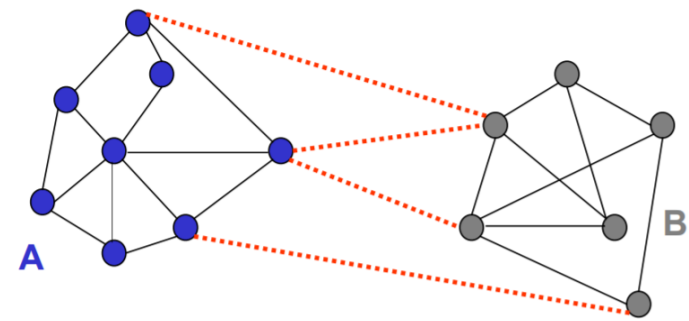
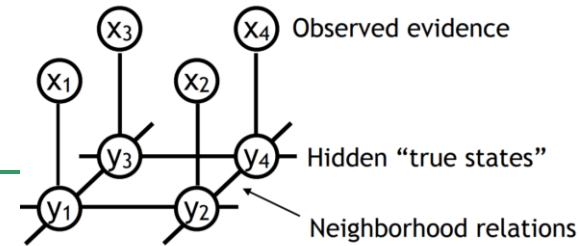


Image segmentation



□ How? -> Computer Vision's era

■ Segmentation as energy minimization

□ Main idea

- Markov Random Fields (MRFs) and Conditional Random Fields (CRFs)
 - Rich probabilistic model for images
 - Built in local, modular way - Get global effects from only learning/modeling local ones
 - After conditioning, get a Markov Random Field (MRF)

□ Algorithms

- Grab-Cut (2004)

□ See

- Boykov, Y., Veksler, O., & Zabih, R. (2001). Fast approximate energy minimization via graph cuts. *IEEE Transactions on pattern analysis and machine intelligence*, 23(11), 1222-1239.

<http://luthuli.cs.uiuc.edu/~daf/courses/Opt-2017/Combinatorialpapers/00969114.pdf>

- + Very powerful, get global results by defining local interactions
- + Very general
- + Rather efficient
- - Only works for sub modular energy functions (binary)
- - Only approximate algorithms work for multi-label case

Image segmentation

□ How? ->Computer Vision's era

■ Region-based segmentation (low-level methods)

□ Main idea

- rely mainly on the assumption that the neighboring pixels within one region have similar values.

□ Algorithms

- Thresholding (Otsu's algorithm), Region growing (GrowCut)

□ See

- <https://im.snibgo.com/growcut.htm>

□ + simple, fast,

□ - doesn't work if there are overlapped gray levels in image

■ Machine learning algorithms (high-level methods)

□ + simple, general

□ - high training time

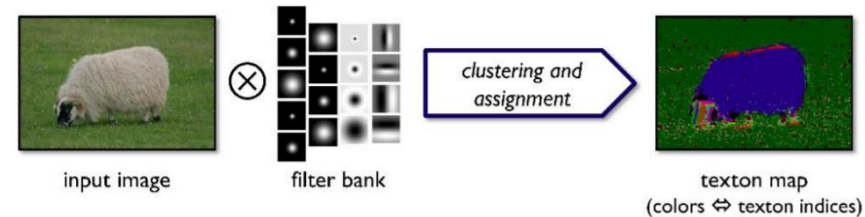
Image segmentation

□ How? -> Computer Vision's era

■ Machine learning algorithms

□ Before deep learning

- CRF + pixels/superpixels

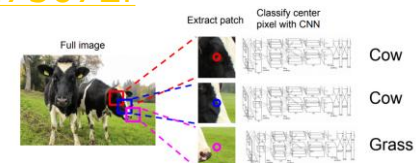


- Jamie Shotton

<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=10.3a26b7066269523698278314ebf1143a175072f>

- CRFs <https://pub.ist.ac.at/~chl/papers/>

- Sliding window



- <http://yann.lecun.com/exdb/publis/pdf/farabet-pami-13.pdf>

- https://ronan.collobert.com/pub/matos/2014_scene_icml.pdf

□ Deep learning era

- Unet, Unet++, U2net &co

- see <https://causlayer.o>

- SegNet

- DeepLab

- FCN

- DenseNet

- ...

- Please check

- <https://github.com/mrgloom/awesome-semantic-segmentation>

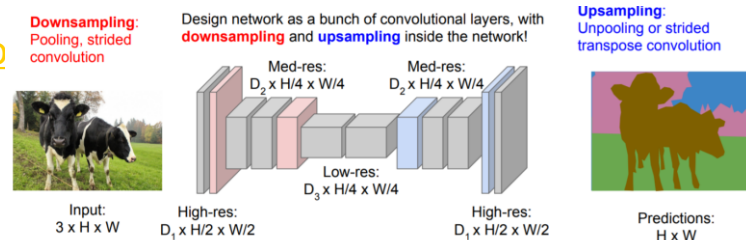


Image segmentation

- How? -> Computer Vision's era
 - Deep learning algorithms
 - Main idea
 - Extract features -> encoding
 - Decoding and classify pixels
 - Algorithms
 - Fully Convolutional Networks
 - Convolutional Networks with Graphical Models (CRFs and MRFs)
 - Multi-scale and Pyramid Network based models
- + simple, general
- - high training time

Image segmentation

□ Problem (modern formulation)

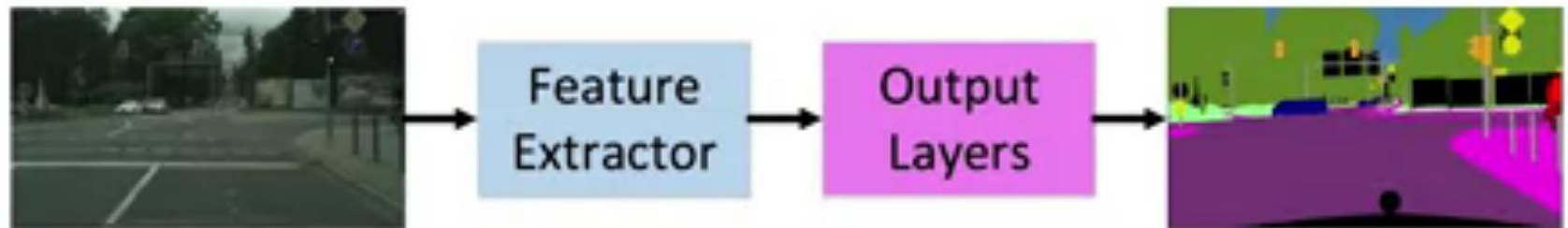
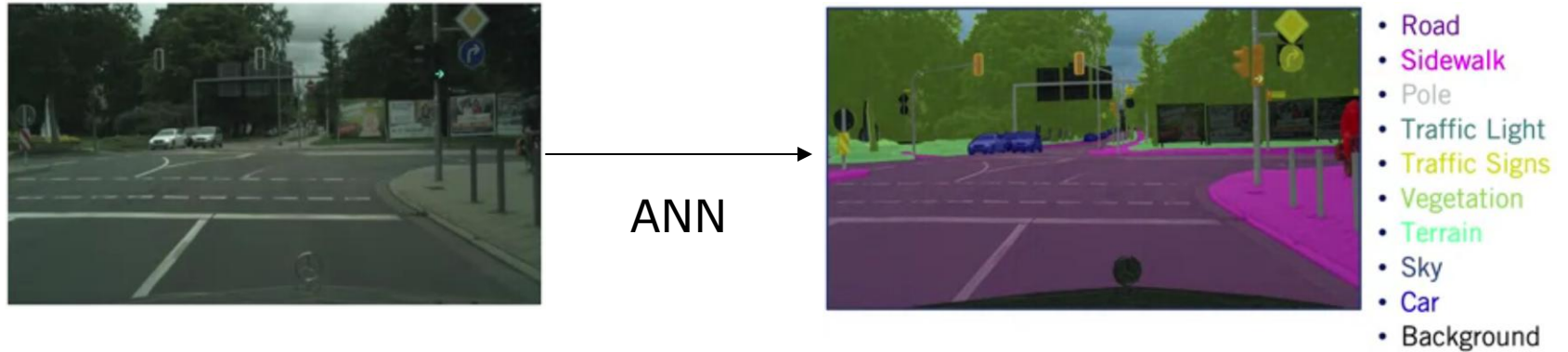


Image segmentation

□ Semantic segmentation



Input

segmented →

- 1: Person
- 2: Purse
- 3: Plants/Grass
- 4: Sidewalk
- 5: Building/Background

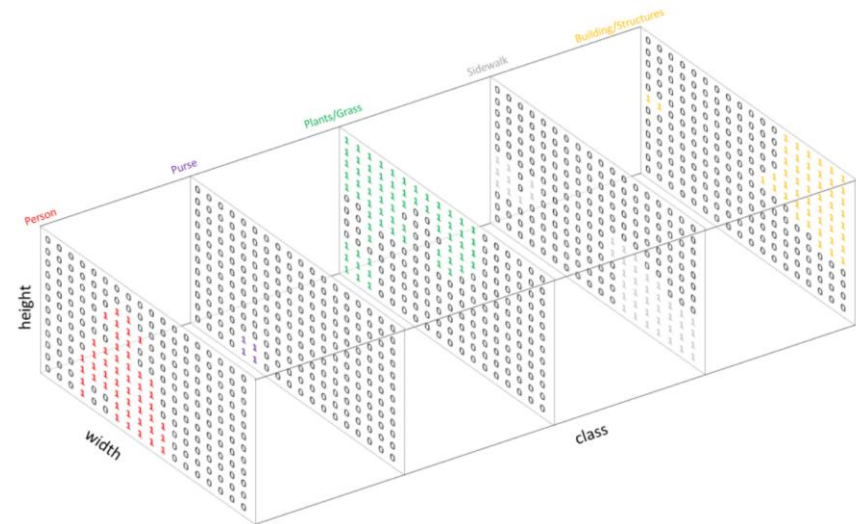


Image segmentation

□ Problem -> feature extraction

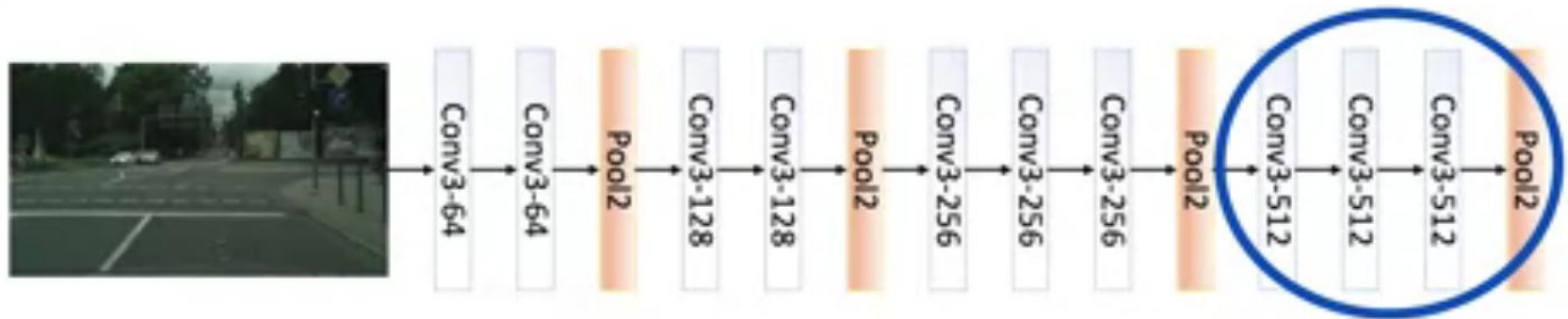
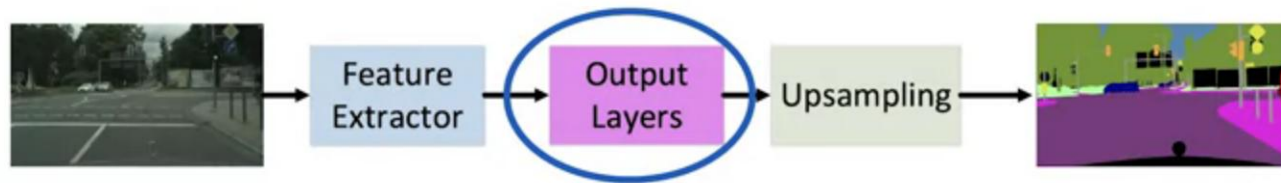


	Image	Conv1	Conv2	Conv3	Conv4
Width	M	M/2	M/4	M/8	M/16
Height	N	N/2	N/4	N/8	N/16
Depth	3	64	128	256	512

Image segmentation

□ Up-sampling



□ Learning same resolution feature maps

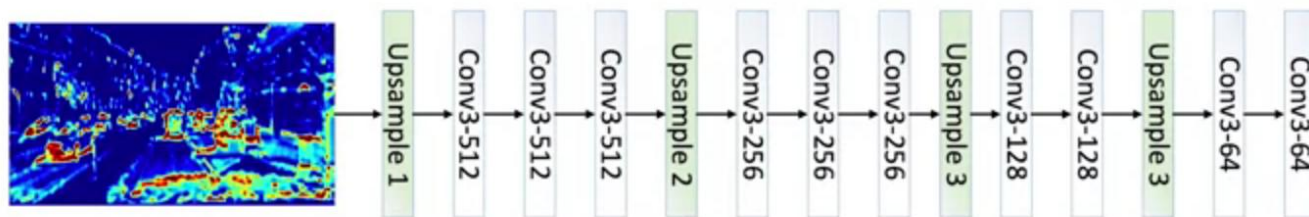
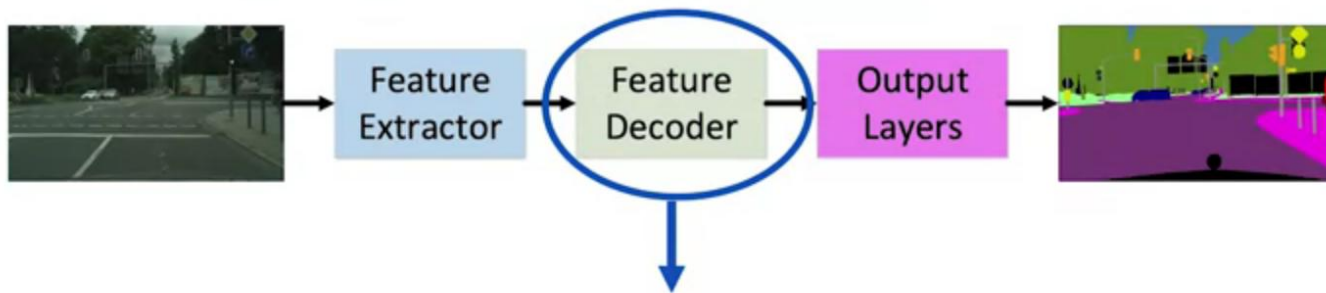
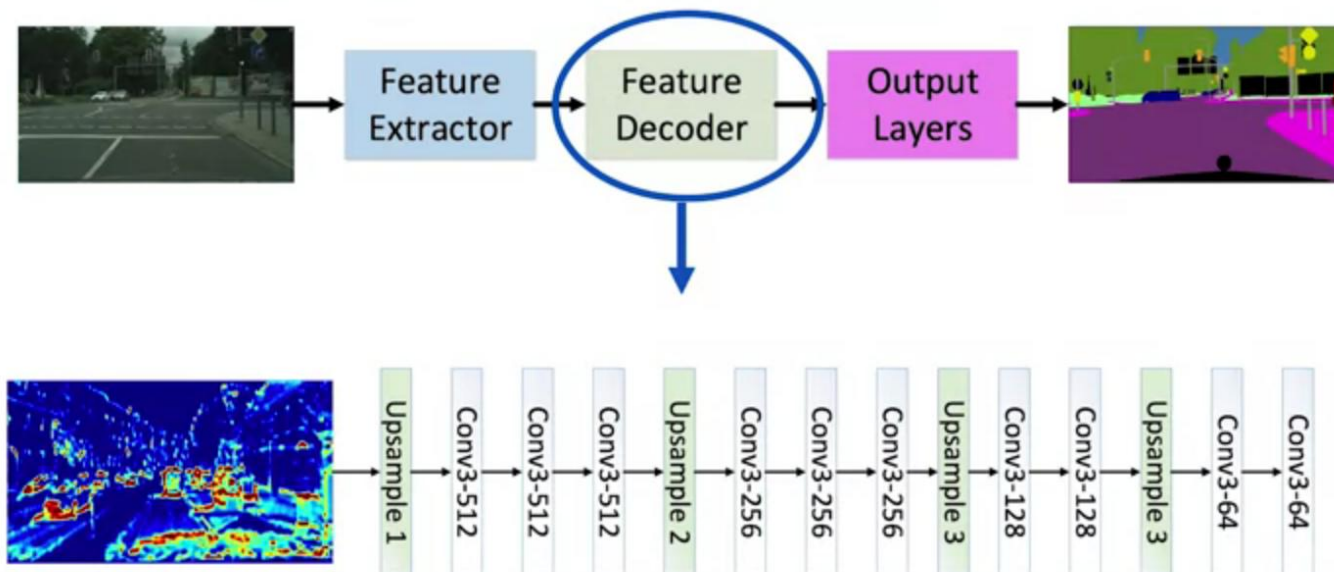


Image segmentation

□ Learning same resolution feature maps



	Feature Map	Deconv1	Deconv2	Deconv3	Deconv4
Width	M/16	M/8	M/4	M/2	M
Height	N/16	N/8	N/4	N/2	N
Depth	512	512	256	128	64

Image segmentation

□ Output computation

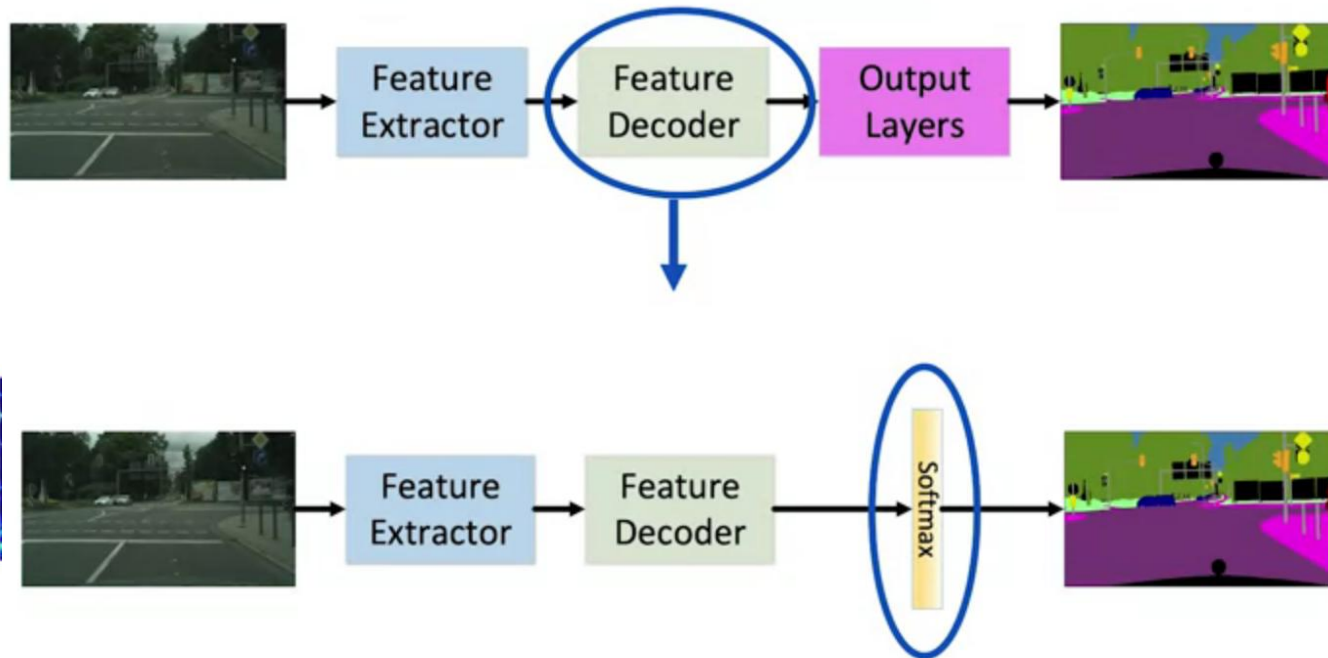


Image segmentation

- Fully convolutional networks
 - Feature extraction by convolutions (down-sampling / encoder path)
 - Extract and interpret the context (what?)
 - Segmentation map by recovering spatial information by convolutions (up-sampling / decoder part)
 - Enable precise location (where?)
 - Transform FC layers from a classification architecture into 1/more convolutions (deconvolutions or transposed convolutions) -> up-sampling
 - Skip connections
 - Recover the fine-grained spatial information lost in pooling or down-sampling layers
 - Merge (concatenate or sum) more feature maps from the down-sampling path with feature maps from the up-sampling path
 - Helps combining context information with spatial information

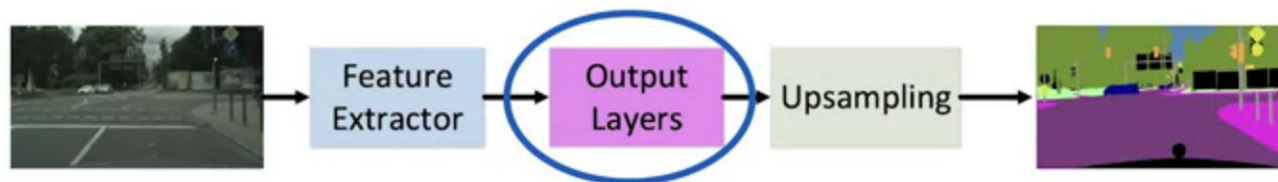


Image segmentation

□ Fully convolutional networks

■ Most common architectures

□ FCN

- https://www.cv-foundation.org/openaccess/content_cvpr_2015/papers/Long_Fully_Convolutional_Networks_2015_CVPR_paper.pdf
- Non-symmetric paths
 - #feature maps = #classes (down-sampling)
 - Up-samples only once (one layer for decoding + bilinear interpolation)
- Skip connections by sum

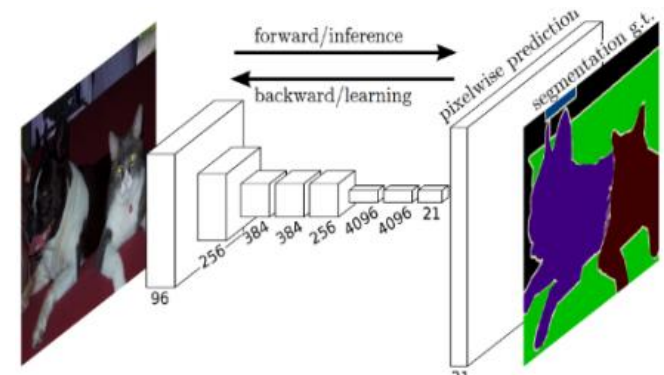


Image segmentation

□ Fully convolutional networks

■ Most common architectures

□ Unet

- <https://arxiv.org/pdf/1505.04597.pdf>
- a symmetric architecture
 - Larger #feature maps
 - Multiple up-sampling layers (=> learnable weight filters for interpolation)
- Skip connections by concatenation

□ SegNet

- <https://arxiv.org/pdf/1511.00561.pdf>
- Similar to Unet

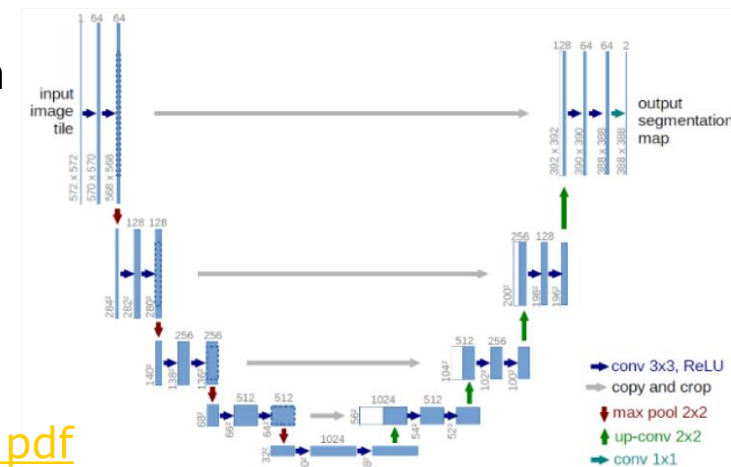


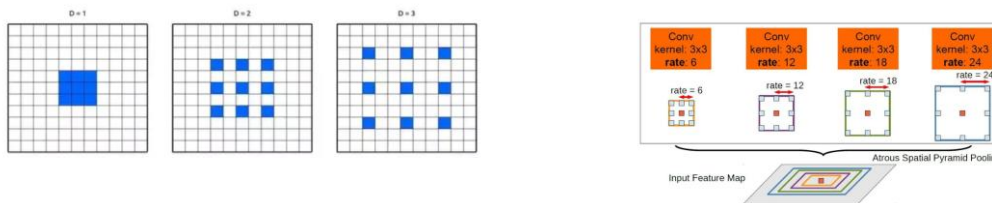
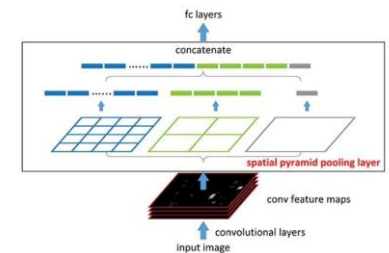
Image segmentation

□ Fully convolutional networks

■ Most common architectures

□ DeepLab

- <https://github.com/tensorflow/models/tree/master/research/deeplab>
- New elements
 - Spatial pyramid pooling
 - Dilated (atrous) convolutions
 - Depthwise separable convolutions
 - Improving outputs with CRF



□ DenseNet and many others

- <https://github.com/mrgloom/awesome-semantic-segmentation>

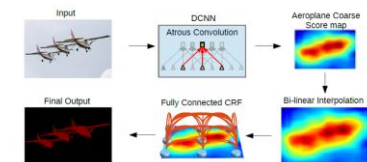


Image segmentation

□ Unet / SegNet vs DeepLab

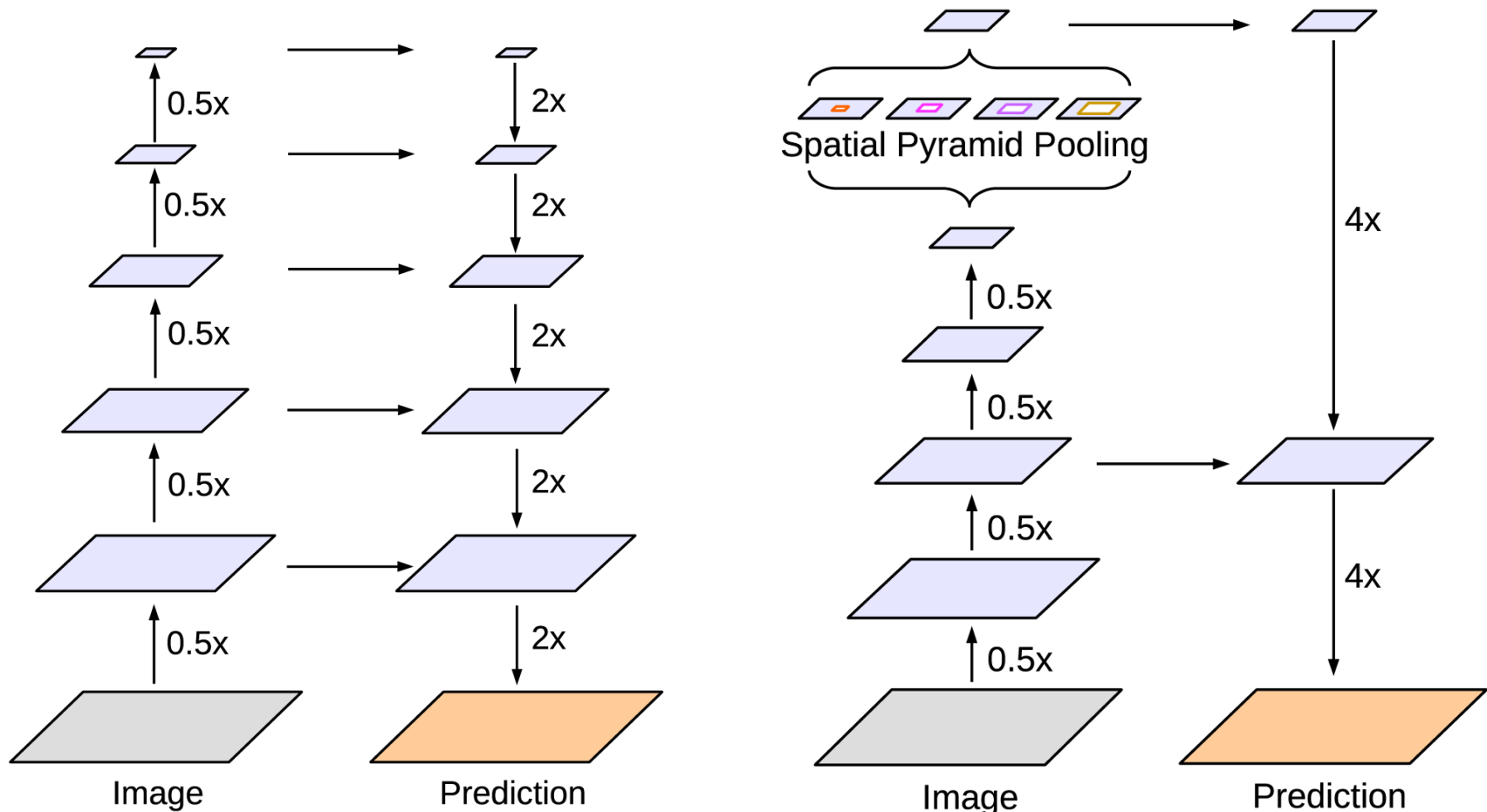


Image segmentation

□ Mask R-CNN

- <https://arxiv.org/pdf/1703.06870.pdf>

- Similar to Faster R-CNN, but predict masks as well as BBs

- a Fully CNN (on top of feature map) for determining a binary mask (object or not) for each RoI

- RoI Alignment -> bilinear interpolation

- $\text{Loss} = \text{Loss}(\text{classific}) + \text{Loss}(\text{bb}) + \text{Loss}(\text{mask})$

- $\text{Loss}(\text{mask}) = \text{cross-entropy}$

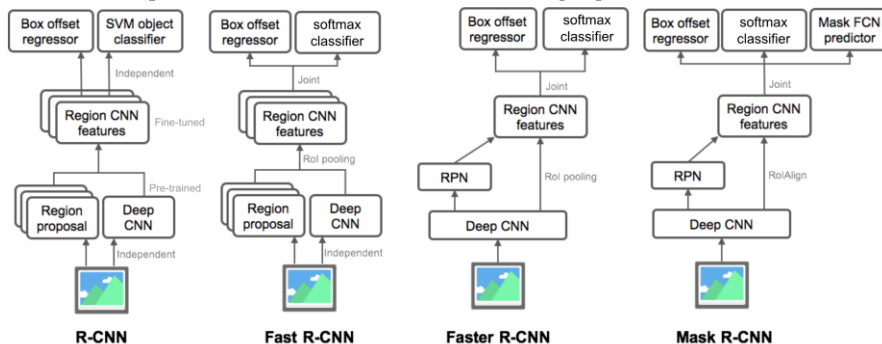
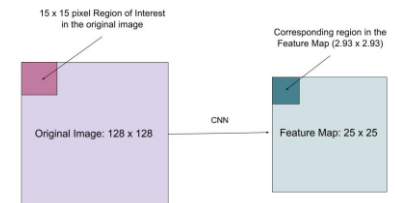
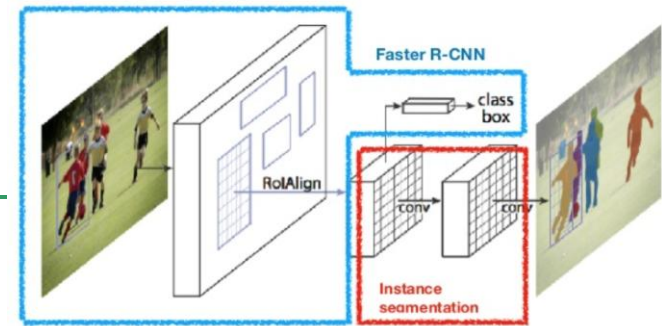


Image segmentation

□ Detectron

■ Feature extraction

- Feature pyramid network
- Different backbones (ResNet)

■ Proposal generator

- Region proposal network

■ Target tasks

- BB prediction
- BB classification
- Pixel-level classification inside a BB (segmentation)

■ $\text{Loss} = \text{Loss}(\text{classific}) + \text{Loss}(\text{bb}) + \text{Loss}(\text{mask})$

- $\text{Loss}(\text{mask}) = \text{cross-entropy}$
- Focal loss

■ Non-local NN <https://arxiv.org/pdf/1711.07971.pdf>

- Long-range dependencies
 - Recurrent operations (repeated convolutions = local neighbourhood)
 - Non-local operations
 - Mean of all positions of an input = a very large receptive field
 - Self-attention (machine translation)
 - CRF (graphical models)

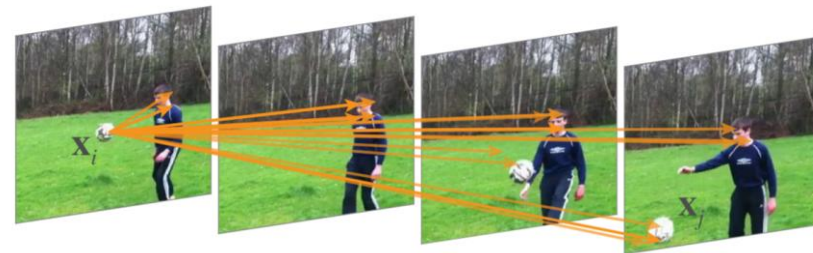


Image segmentation

□ YOLACT (You Only Look At CoefficientTs)

- <https://github.com/dbolya/yolact>
- <https://arxiv.org/pdf/1904.02689.pdf>

□ Vision Transformers (ViT)

- reducing architecture complexity
- exploring scalability and training efficiency

■ **An Image is Worth 16x16 Words**

- <https://arxiv.org/pdf/2010.11929.pdf>
- <https://ai.facebook.com/research/publications/end-to-end-object-detection-with-transformers>
- **NLP transformers** <http://jalammar.github.io/illustrated-transformer/>
- https://github.com/google-research/vision_transformer

Image segmentation

More details

- <https://arxiv.org/pdf/2001.05566.pdf>
- <https://heartbeat.fritz.ai/a-2019-guide-to-semantic-segmentation-ca8242f5a7fc>
- <https://paperswithcode.com/sota/instance-segmentation-on-coco>
- <https://link.springer.com/article/10.1007/s13735-020-00195-x>

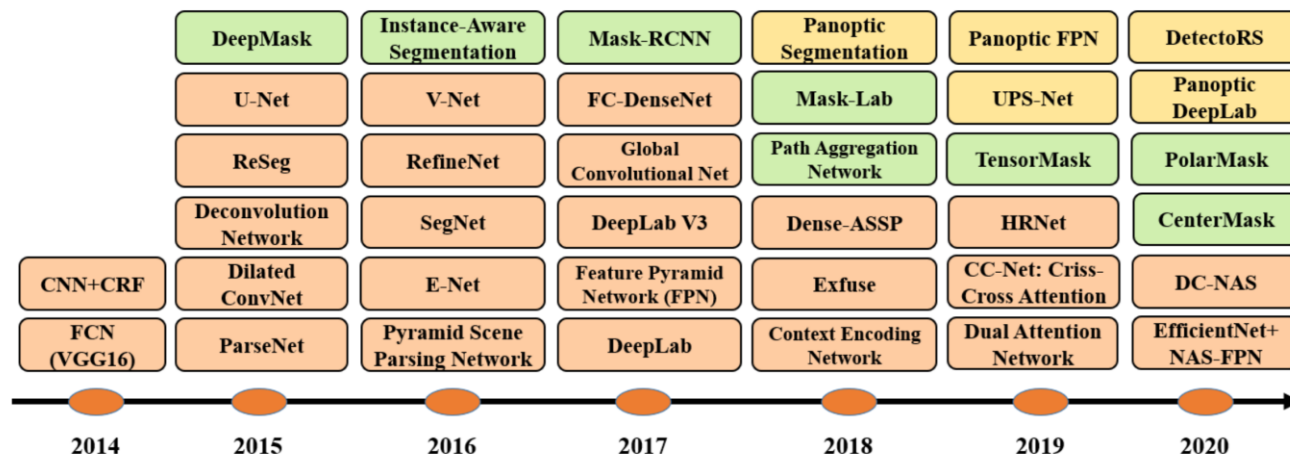


Fig. 32. The timeline of DL-based segmentation algorithms for 2D images, from 2014 to 2020. Orange, green, andn yellow blocks refer to semantic, instance, and panoptic segmentation algorithms respectively.

Image segmentation

□ Segmentation

- partitioning an image into meaningful segments, which share a common representation.
- Dense pixel prediction -> it classifies each pixel into one of a few classes

□ Semantic segmentation

- Segment all interest objects (by different classes = semantic classes)

□ Instance segmentation

- Segment all interest objects (by different classes = semantic classes)
- Predict an instance label for each object of interest

□ Panoptic segmentation

- Instance segmentation of all interest objects (by classes) and the background